

Day 10 – Advanced Production Shell Scripting

Objective:

Learn production-grade shell scripting techniques including argument handling, configuration files, lock files, and script structuring.

1. Argument Handling:

Scripts can accept dynamic input using positional parameters.

Example: `./script.sh 80 logfile.log`

Use `$1`, `$2` to capture values.

2. Configuration File:

Instead of hardcoding values, use `config.conf`.

Load using: `source config.conf`

3. Lock File:

Prevents multiple script executions.

Example: `LOCK_FILE=/tmp/script.lock`

4. Trap for Cleanup:

Ensures lock file removal even if script fails.

Example: `trap "rm -f $LOCK_FILE" EXIT`

5. Log Rotation:

Rotate logs when size exceeds limit.

6. Final Script Features:

- Disk monitoring
- Service monitoring
- Connectivity checks
- Logging system
- Alert system
- Config-based setup
- Lock file handling

7. Cron Automation:

```
*/10 * * * * /path/to/script.sh
```

Best Practices:

- set -euo pipefail
- Validate inputs
- Use config files
- Implement logging
- Handle errors

Conclusion:

Day 10 marks transition to production-ready DevOps scripting.